

# MULTIPLE CHOICE QUESTIONS ANSWER KEY

Questions with Correct Options Marked

**Q1. A perceptron has weights [0.2, -0.1], receives  $x = [1, 2]$ , predicts 0 for target 1, and uses learning rate 0.1. After one perceptron update, what are the new weights?**

• [0.1, -0.3]

✓ [0.3, 0.1]

• [0.3, -0.2]

• [0.2, 0.1]

**Q2. A multi-layer network contains several affine layers but no nonlinear activation between them. Which statement best explains its resulting decision capacity?**

✓ It behaves like one affine transformation despite the added depth

• It gains nonlinear boundaries because each layer has separate weights

• It behaves like a kernel classifier once more than two layers are stacked

• It gains piecewise-linear boundaries because biases differ across layers

**Q3. A deep MLP uses ReLU activations and its initial activations become excessively large as depth increases. Which initialization is most directly intended to control variance for ReLU layers?**

• Large constant initialization

✓ He-style variance-scaled initialization

• All-zero weight initialization

• Uniform initialization with identical weights

**Q4. In a deep network using sigmoid units, many hidden neurons operate at very large positive or negative pre-activations and early layers learn extremely slowly. Which mechanism best accounts for this behaviour?**

• Softmax competition increases early-layer gradients

✓ Sigmoid saturation produces very small local derivatives

• Mini-batch averaging reverses the sign of most gradients

• Cross-entropy forces hidden activations towards exact zero

**Q5. A dataset has 1,000 training samples and mini-batch gradient descent uses a batch size of 100. Ignoring an incomplete final batch, how many parameter updates occur in one epoch?**

• 5

✓ 10

• 100

• 1,000

**Q6. A softmax classifier produces logits [7, 5, 4]. Before applying softmax, the same constant is subtracted from every logit for numerical stability. What happens to the output probabilities?**

✓ **They preserve exactly the same class probabilities**

- They preserve only the highest-probability class
- They become uniformly distributed across the classes
- They change proportionally to the subtracted constant

**Q7. During full-batch gradient descent, training loss repeatedly crosses a narrow minimum and alternates between two nearby but higher values. Which adjustment most directly addresses this optimisation behaviour?**

- Increase the learning rate to cross the valley faster

✓ **Reduce the learning rate to make smaller updates near the minimum**

- Replace the loss with softmax while keeping the same step size
- Increase the batch size beyond the complete training set

**Q8. A  $31 \times 31$  feature map is processed by a  $3 \times 3$  convolution with stride 2 and padding 1. What is the spatial size of the output feature map?**

- $15 \times 15$

✓  **$16 \times 16$**

- $17 \times 17$
- $18 \times 18$

**Q9. A  $32 \times 32$  feature map passes through  $2 \times 2$  max pooling with stride 2 and no padding. Which description of the output is correct?**

✓  **$16 \times 16$  with no trainable pooling weights**

- $16 \times 16$  with one trainable weight per window
- $31 \times 31$  with no trainable pooling weights
- $31 \times 31$  with one trainable weight per window

**Q10. A very deep plain CNN shows higher training error after adding more convolutional layers. Which architectural modification most closely targets the optimisation problem addressed by ResNet?**

✓ **Insert identity shortcut paths around groups of layers**

- Replace all  $3 \times 3$  filters with larger  $7 \times 7$  filters
- Move every pooling layer ahead of its convolutional block
- Replace convolutional blocks with additional dense layers

**Q11. A pre-trained image classifier was trained with a specific channel-wise mean and standard deviation. Which preprocessing choice is most appropriate when reusing its backbone?**

✓ **Apply the same normalization expected during pre-training**

- Normalize each image independently to zero mean and unit range
- Use only min-max scaling because the backbone is already trained
- Skip normalization so pre-trained convolution filters remain unchanged

**Q12. In a medical image dataset, small rotations preserve the diagnosis but horizontal flipping can exchange left- and right-sided findings. Which augmentation policy is most defensible?**

✓ **Use modest random rotations but avoid horizontal flips**

- Use horizontal flips but avoid all small rotations
- Use both transformations with equal probability
- Use neither transformation and only duplicate the images

**Q13. A vision application needs the class label and bounding box of every visible object, but it does not require a pixel-level mask. Which task best matches the requirement?**

- Image classification

✓ **Object detection**

- Semantic segmentation
- Image reconstruction

**Q14. A small target dataset is visually similar to the data used to pre-train a CNN. Training a newly replaced classifier head has stabilised. Which next step is most suitable if additional adaptation is required?**

✓ **Unfreeze selected upper backbone layers and fine-tune with a smaller learning rate**

- Reinitialise the complete backbone and keep only the classifier weights
- Freeze the classifier and update only the earliest convolutional layers
- Increase the learning rate and update every layer from the pre-trained values

**Q15. Two neighbouring image regions have the same semantic class, but an application must distinguish them as separate physical objects. Which output representation is required?**

- One class label for the whole image
- One semantic class map shared by all objects

✓ **Separate instance masks for the individual objects**

- One feature vector for each convolutional channel

**Q16. In self-attention, a query at one token has a much larger similarity score with the key of token  $j$  than with other keys. What is the most direct consequence after softmax?**

✓ **Token  $j$ 's value receives a larger contribution weight**

- Token  $j$ 's query replaces the current token's query
- Token  $j$ 's key is added directly to the output vector
- Token  $j$ 's positional encoding is removed from the sequence

**Q17. An encoder processes batches of sequences padded to a common length. Which masking choice prevents padded positions from influencing representations of real tokens?**

- Mask only positions occurring after the current token

✓ **Mask the key positions corresponding to padding tokens**

- Mask every token having the same embedding as padding tokens
- Mask the query positions corresponding to all non-padding tokens

**Q18. Why can multi-head attention represent relationships that may be harder for a single head of the same total dimension to capture?**

✓ **Different heads learn separate projection subspaces before their outputs are combined**

- Different heads use identical projections but apply different softmax functions
- Each head attends to a fixed non-overlapping part of the sequence
- Each head receives a different positional encoding length for the same tokens

**Q19. A Transformer uses layer normalization within its blocks. Which normalization behaviour is most consistent with layer normalization in sequence models?**

✓ **Normalize hidden features of each token independently of other batch examples**

- Normalize each feature using statistics collected across the complete dataset
- Normalize attention heads using one shared statistic across the training batch
- Normalize only the final class probabilities after the softmax layer

**Q20. Within a standard Transformer block, how is the feed-forward sublayer applied to a sequence after attention?**

✓ **The same feed-forward network is applied independently at each sequence position**

- A different feed-forward network is trained for every sequence position
- The feed-forward network mixes all token positions through one dense matrix
- The feed-forward network is applied only to positions selected by the largest attention score

**Q21. Two sequences contain exactly the same token embeddings but in different orders. Which addition allows a self-attention model to distinguish their positions before attention is computed?**

✓ **Position-dependent encodings added to token representations**

- A larger vocabulary projection applied after the final layer
- A shared dropout mask applied to every token embedding
- A larger number of value vectors with unchanged token representations

**Q22. A Vision Transformer splits a  $192 \times 192$  image into non-overlapping  $16 \times 16$  patches and prepends one class token. What sequence length enters the encoder?**

• 144

✓ **145**

- 192
- 193

**Q23. What is the main purpose of shifted windows in a Swin Transformer compared with repeatedly using the same fixed local windows?**

✓ **Allow information exchange across neighbouring window boundaries while retaining local attention**

- Convert every local window into one global attention matrix at each layer
- Remove the need for patch embeddings by shifting raw pixels directly
- Keep tokens permanently isolated inside the same local window through all layers

**Q24. During GPT-style decoder-only pre-training, what prediction objective is applied at each eligible token position?**

✓ **Predict the next token from the preceding context**

- Predict the preceding token from the following context
- Predict a masked token using unrestricted bidirectional context
- Predict the complete sequence label from the first token only

**Q25. An LLM is given only a task instruction and a new input, with no worked examples in the prompt. Which prompting setup is being used?**

✓ **Zero-shot prompting**

- Few-shot prompting
- Chain-of-thought prompting
- Parameter-efficient fine-tuning

**Q26. A prompt includes two labelled examples before asking the model to solve a new case in the same format. Which prompting technique is primarily being used?**

- Zero-shot prompting

✓ **Few-shot prompting**

- Full fine-tuning
- Retrieval augmentation

**Q27. A team wants separate lightweight task adaptations while keeping one shared LLM checkpoint unchanged. Which approach most closely matches LoRA-based PEFT?**

✓ **Train low-rank update matrices while freezing the base model weights**

- Duplicate and fully fine-tune every base-model weight for each task
- Update only token embeddings while permanently removing attention projections
- Retrain the language model from random initialization for each task

**Q28. In a basic RAG pipeline, what should normally happen immediately before the retrieved text is supplied to the generator?**

✓ **Represent the query and retrieve semantically similar stored chunks**

- Fine-tune the generator on every document returned by the user
- Replace document embeddings with the generator's output logits
- Regenerate the entire vector store from the language model response

**Q29. A RAG system increases overlap between neighbouring document chunks. Which trade-off is most directly introduced?**

✓ **Better preservation of boundary context with greater retrieval redundancy**

- Lower preservation of boundary context with fewer indexed vectors
- Better preservation of boundary context with no change in index redundancy
- Lower preservation of boundary context with greater semantic separation

**Q30. An LLM application must search a database, use a calculator on the retrieved values, and then decide whether another tool call is required. Which design characteristic makes this workflow agentic?**

✓ **The next action is selected using the current state and previous tool observations**

- All tool calls are fixed before any tool result is available
- The model produces one response without observing tool outputs
- The calculator replaces the language model as the planning component

**Q31. An MLP receives  $x_1 = 1$  and  $x_2 = 2$ . Its hidden units are  $h_1 = \text{ReLU}(2x_1 - x_2 + 1)$  and  $h_2 = \text{ReLU}(-x_1 + x_2 - 0.5)$ . If the output is  $o = 1.5h_1 + 2h_2 - 0.5$ , what is  $o$ ?**

• 1.5

✓ 2.0

• 2.5

• 3.0

**Q32. A linear neuron uses  $z = w \cdot x$  with  $w = 0.5$  and  $x = 4$ . For target  $y = 1$  and loss  $L = 0.5(z - y)^2$ , one gradient-descent step uses learning rate 0.1. What is the updated weight?**

✓ 0.1

• 0.3

• 0.4

• 0.9

**Q33. A fully connected MLP has 10 input features, one hidden layer with 6 neurons, and 3 output neurons. Every neuron in the hidden and output layers has a bias. How many trainable parameters are present?**

• 81

• 84

✓ 87

• 90

**Q34. A convolutional layer receives 8 input channels and uses 24 filters of size  $3 \times 3$ , with one bias per filter. How many trainable parameters does the layer contain?**

• 1,728

✓ 1,752

• 1,776

• 1,800

**Q35. A  $64 \times 64$  feature map passes through a  $5 \times 5$  convolution with stride 2 and padding 2, followed by  $2 \times 2$  max pooling with stride 2. What is the final spatial size?**

•  $15 \times 15$

✓  $16 \times 16$

•  $17 \times 17$

•  $32 \times 32$

**Q36. A pre-trained CNN underfits a small target dataset when the whole backbone is frozen, but rapidly overfits when the entire backbone is unfrozen. Which next strategy provides the best balance?**

✓ **Unfreeze only upper backbone blocks, use a smaller learning rate, and retain suitable augmentation**

- Keep every backbone block frozen, increase only the head learning rate, and remove augmentation
- Unfreeze the entire backbone, increase the learning rate, and reduce regularisation
- Reinitialise the entire backbone, freeze the classifier head, and train from scratch

**Q37. For one attention query, the scaled logits for two keys differ by 0.707 in favour of key 2. Using  $e^{0.707} \approx 2.03$ , what softmax weight is assigned approximately to key 2?**

- 0.33
- 0.50

✓ **0.67**

- 0.73

**Q38. A Transformer has  $d_{\text{model}} = 768$ , 12 equal-sized attention heads, and sequence length 100. Which pair correctly gives the dimension per head and the attention-score matrix size per head?**

✓ **64 and  $100 \times 100$**

- 64 and  $100 \times 64$
- 768 and  $100 \times 100$
- 768 and  $100 \times 64$

**Q39. The sequence length entering full self-attention increases from 128 to 256 while head dimension and number of heads remain unchanged. By what factor does the number of attention-score entries increase?**

- 2

✓ **4**

- 8
- 16

**Q40. A self-attention layer receives a sequence of length 40 with  $d_{\text{model}} = 512$  and 8 equal heads. Which set of shapes is consistent with standard multi-head attention?**

✓ **Per-head Q:  $40 \times 64$ , per-head scores:  $40 \times 40$ , concatenated output:  $40 \times 512$**

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- Per-head Q:  $40 \times 64$ , per-head scores:  $64 \times 64$ , concatenated output:  $40 \times 512$



**Q41. LoRA is applied to a  $2,048 \times 1,024$  weight matrix using rank  $r = 4$ . If only the two low-rank matrices are trainable, how many trainable parameters are added for this weight?**

• 8,192

✓ 12,288

• 16,384

• 2,097,152

**Q42. A prompting experiment gives correct content but inconsistent output structure across inputs. The model weights must remain unchanged. Which modification most directly improves format consistency while staying within prompt engineering?**

✓ Provide a precise output template together with a few representative formatted examples

- Increase the number of trainable adapter parameters without changing the prompt
- Replace retrieval embeddings while keeping the prompt unchanged
- Continue decoder pre-training on unrelated text before each request

**Q43. During decoder-only pre-training, the model assigns probabilities 0.5 and 0.25 to the correct next tokens at two positions. Using natural-log loss and  $\ln(2) \approx 0.693$ , what is the mean negative log-likelihood?**

• 0.693

✓ 1.040

• 1.386

• 2.079

**Q44. A RAG system often misses answers whose supporting sentence is split across two neighbouring chunks, although both source documents are indexed. Which change should be tested first?**

✓ Re-chunk around more coherent boundaries and introduce suitable overlap

- Increase generator temperature while leaving retrieval unchanged
- Fine-tune the generator before checking whether retrieval finds the evidence
- Reduce the vector-store size by removing neighbouring chunks

**Q45. An autonomous agent can read account data and can also execute an irreversible transaction. Which control best supports responsible deployment without removing useful autonomy?**

✓ Allow read-only tools autonomously but require explicit approval before the irreversible action

- Allow every tool autonomously once the initial user request has been received
- Disable tool observations so the model cannot revise its original action plan
- Permit the transaction whenever the model assigns the action a high internal probability